310 Lab Report 2

I started by reading randomint100.txt and printing sum. I started with declaring the filename, newline character, and "Sum: " string in the .data section and reserve space for a buffer, an integer for summing, an integer for the current number, and a string for the value of the sum in the .bss section. The Assembly is built off of system calls where eax = 5 opens the file, eax = 3 reads into the buffer, and from there a loop occurs over the buffer. Each character is verified through conditional checks against '0' and '9'. If it is a digit, it's converted to an integer by offsetting it by '0'; the integer of the current number is obtained by multiplying 10 by the current integer plus the digit. If not a digit, it's a space which indicates the end of a number; the current number is added into the sum, and the current number resets. Once the loop occurs without error—it's done through comparing to 255 to avoid infinite loops—the sum needs to be converted back into a string. This is done by dividing by ten, getting the last digit for reversing, then indexing it into the string until characters are gone, and finally a null character is appended. In order to get the final output, eax = 4 to print the string "Sum: ", eax = 1 to exit, and then the terminal prints the string that contains the sum as the second action. This makes sense because after deconstructing it, I know to get the file, get digits, get the sum, and print it, which covers all aspects of assembly from declaring variables and literals to getting necessary input and output via external system processes like syscall.